# AZ-305 Agenda

Module 01 Design a governance solution

Module 02 Design a compute solution

Module 03 Design a non-relational data storage solution

Module 04 Design a data storage solution for relational data

Module 05 Design a data integration solution

Module 06 Design an application architecture solution

Module 07 Design Authentication and Authorization Solutions

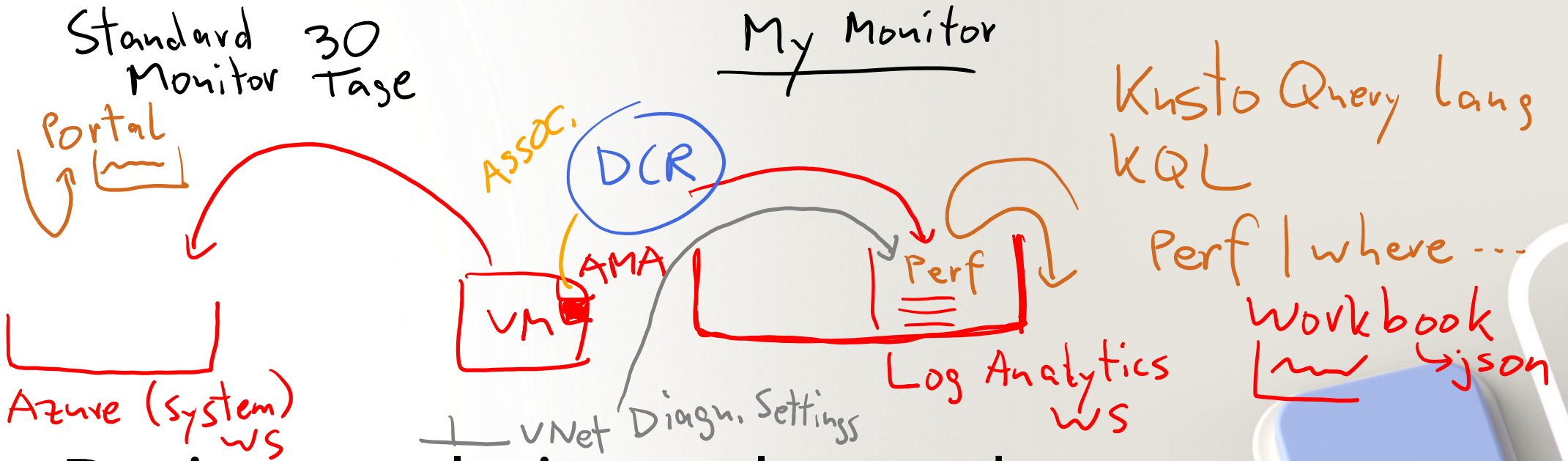
Module 08 Design a solution to log and monitor Azure resources

Module 09 Design a network infrastructure solution

Module 10 Design a business continuity solution

Module 11 Design a migration solution






# Design a solution to log and monitor Azure resources

DCR Data Collection Rule

Sentinel SIEM

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# Learning Objectives

- Design for Azure Monitor data sources
- Design for Log Analytics
- Design for Azure workbooks and Azure Insights
- Design for Azure Data Explorer
- Case study
- Learning recap

AZ-305: Design Identity, Governance, and Monitoring Solutions (25-30%)

Design a Solution for Logging and Monitoring

- Recommend a logging solution
- Recommend a solution for routing logs
- Recommend a monitoring solution

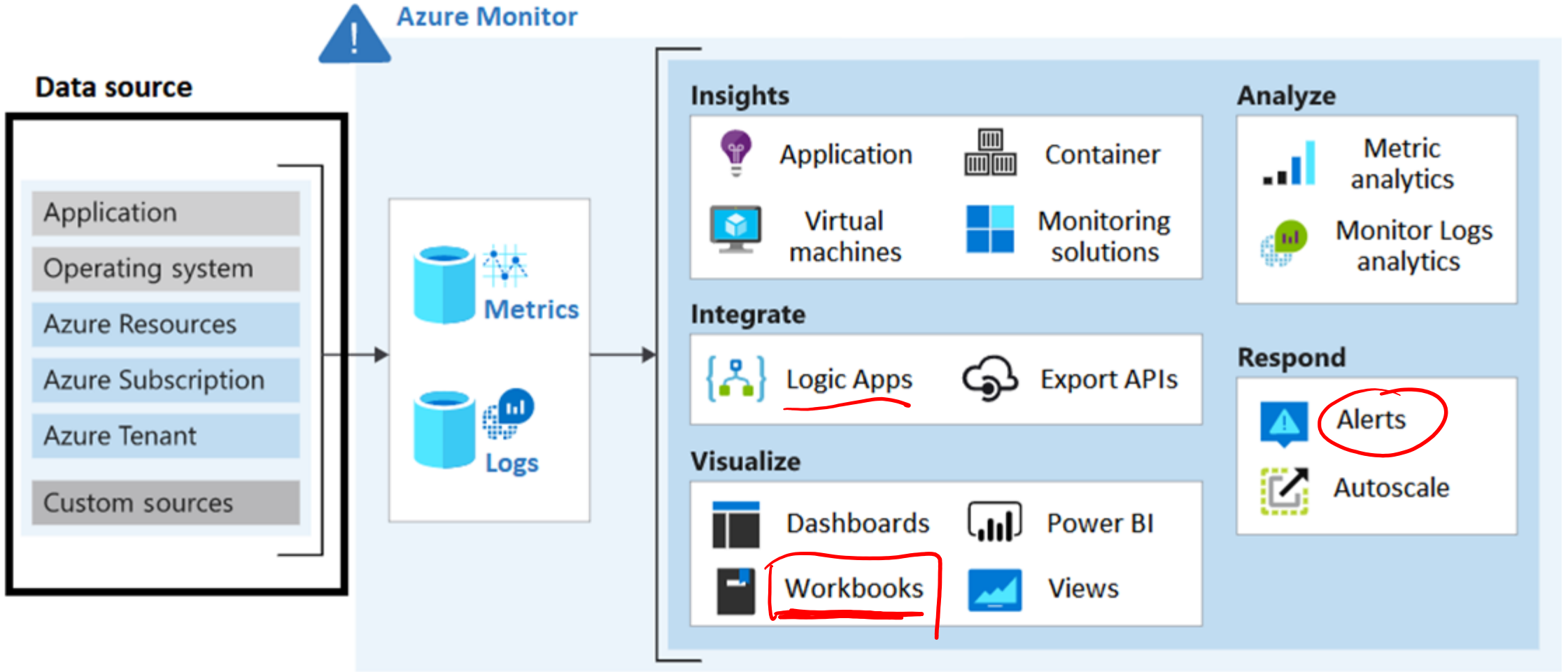
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# Design for Azure Monitor data sources



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# Review Azure Monitor capabilities




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# Identify data sources and access method

## Azure Monitor collects data automatically from a range of components.

- Data tiers go from Azure applications (highest tier) to Azure platform components (lowest tier)
- The method of accessing data from each tier varies – for example, installing an agent
- Each data tier can stream to different external systems
- Prioritize and be deliberate on what data sources you need

- Windows events
- Linux syslog
- Client performance data
- Processes and dependencies (VM Insights)
- Application text logs
- IIS logs
- SNMP traps
- Management pack data (SCOM)

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# Design for Log Analytics



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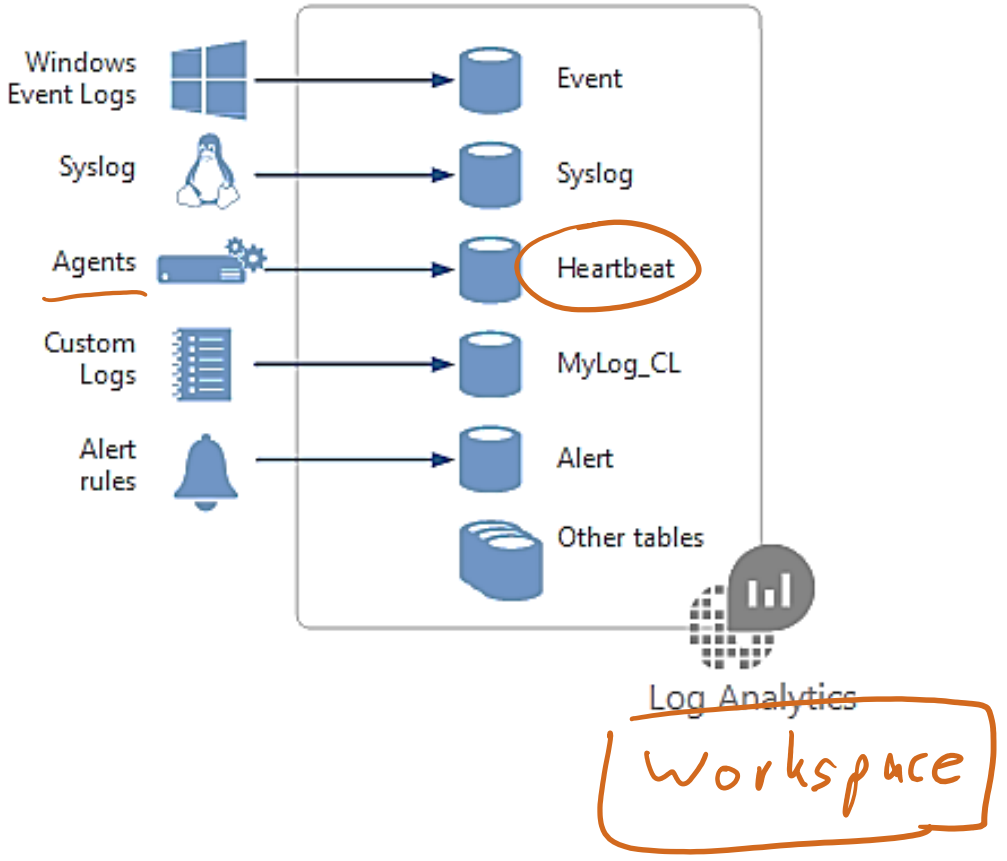
# What is Log Analytics?

## Log Analytics is a service in that helps you collect and analyze data.

- Azure Monitor stores log data in the workspace
- Data in a workspace is organized into tables with properties you can query

A Log Analytics workspace provides:

- A geographic location for data storage.
- Data isolation by granting different users access rights following one of our recommended design strategies.
- Scope for configuration of settings like pricing tier, retention, and data capping.




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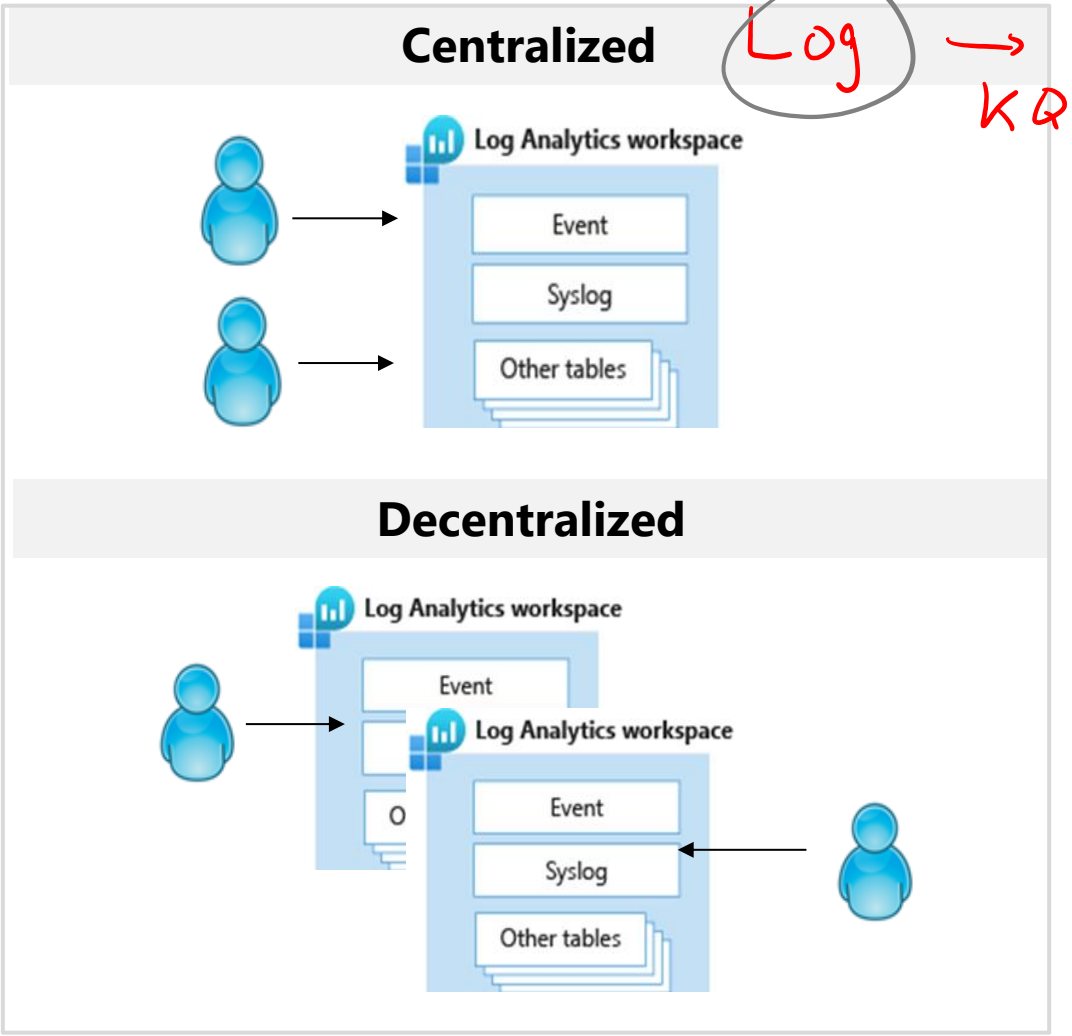
# Considerations for workspace access control

## Workspace deployment models include centralized, decentralized, and hybrid.

**Centralized**: All logs are stored in a central workspace and administered by a single team, with Azure Monitor providing differentiated access per-team.

**Decentralized**: Each team has their own workspace created in a resource group they own and manage, and log data is segregated per resource.

**Hybrid**: Security audit compliance requirements further complicate this scenario because many organizations implement both deployment models in parallel.




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# Considerations for access mode

## The access mode is how a user accesses the workspace and what data they can access.

| Issue | Workspace-context | Resource-context |
|---|---|---|
| **How does the access mode work?** | • You can view all logs in the workspace you have permission to.<br>• Queries in this mode are scoped to all data in all tables in the workspace.<br>• This is the access mode used when logs are accessed with the workspace as the scope. | • When you access the workspace for a particular resource, resource group, or subscription.<br>• You can view logs for only resources in all tables that you have access to.<br>• Queries in this mode are scoped to only data associated with that resource. |
| **Who is each model intended for?** | Central administration | Application teams |
| **What does a user require to view logs?** | Permissions to the workspace | Read access to the resource |
| **What is the scope of permissions?** | Workspace | Azure resource |

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# Design for Azure workbooks and Azure Insights

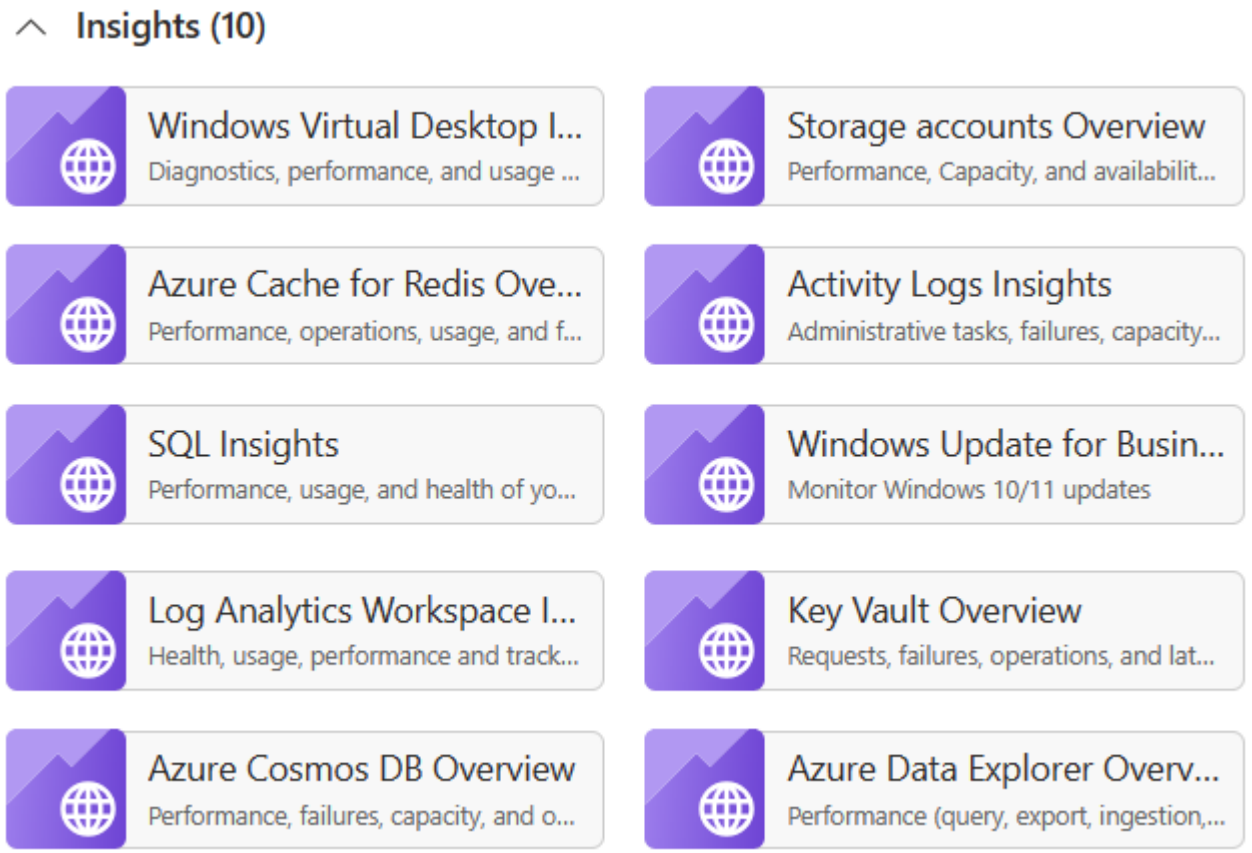


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# Design for Azure Workbooks

## Flexible canvas for data analysis and the creation of rich visual reports

- Tap into multiple data sources and combine them into unified interactive experiences
- Provide insights into the availability, performance, usage, and health of resources
- Enable rich data and insights through composite views and joins




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# Design for Azure Insights

**It's critical to monitor your systems closely to identify any performance problems or attacks before they can affect users. Designing insights as a part of your overall architecture will help identify performance issues.**

## Use Application Insights to:

- Analyze and address issues and problems that affect your application's health and performance.
- Improve your application's development lifecycle.
- Measure your user experience and analyze users' behavior.

## Use Azure Monitor VM insights to:

- View the health and performance of your VMs.
- Monitor your VMs at-scale across multiple subscriptions and resource groups.
- Want a topology view that shows the processes, and network connection details of your VMs and scale sets.

## Use Azure Monitor container insights to:

- View the health and performance of your Kubernetes workloads at-scale across multiple subscriptions and resource groups.
- Want visibility into memory and processor performance metrics from controllers, nodes, and containers.
- Want view and store container logs for real time and historical analysis.

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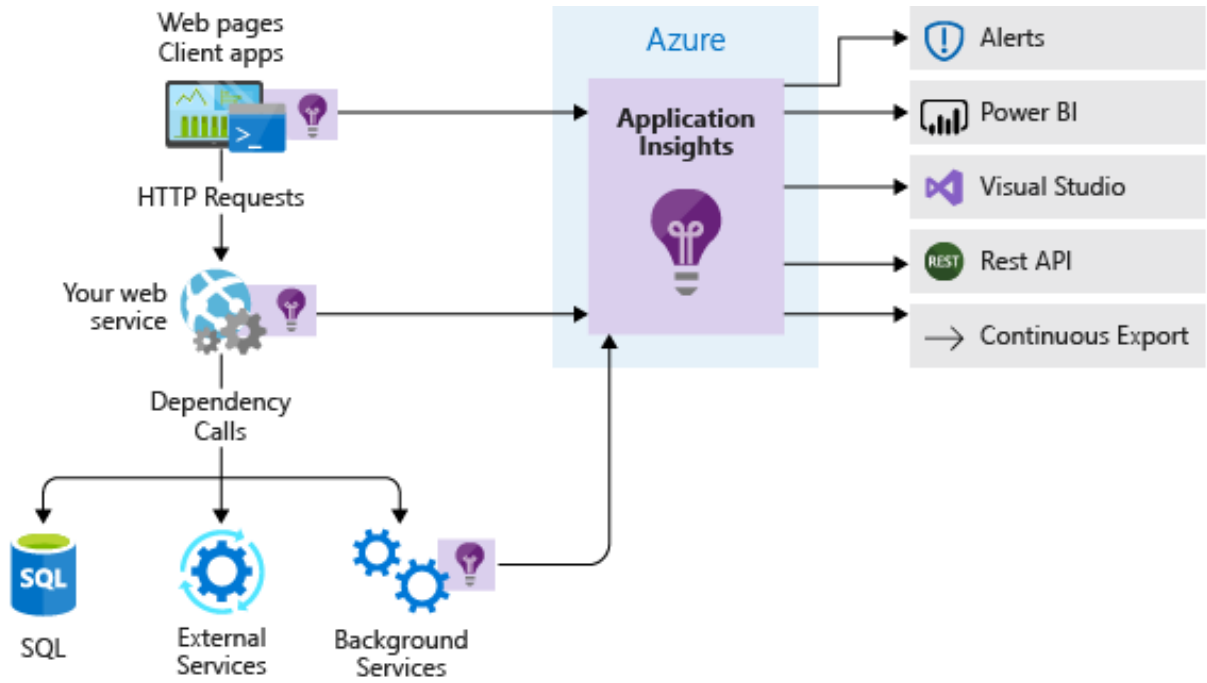
# Select Application Insights

You want to understand how your app is performing and how it's being used.

You need usage information on request rates, response times, and failure rates.

You need transaction diagnostics and performance statistics (client and server).

You want to automatically collect a snapshot of a live application to analyze it at a later stage.




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# Design for Azure Data Explorer

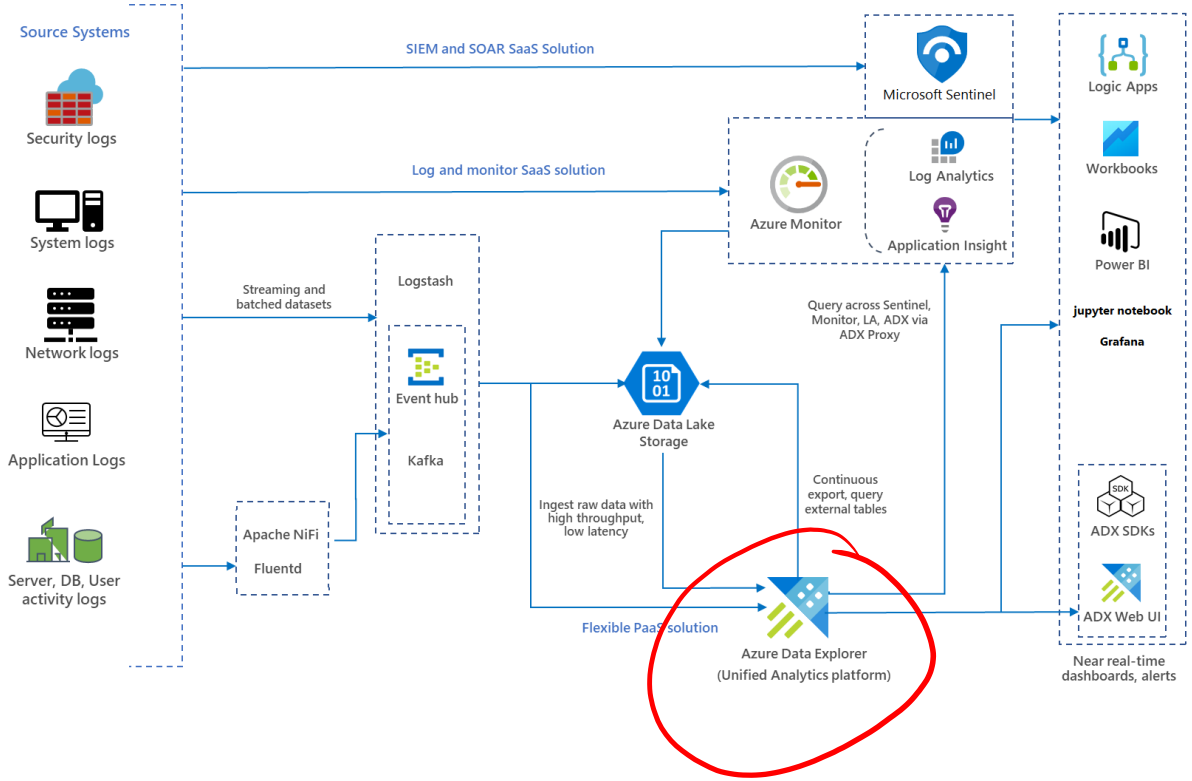


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# When to use Azure Data Explorer

## Azure Data Explorer is a fast and highly scalable data exploration service for log and telemetry data.

- Helps you handle the many data streams emitted by modern software, so you can collect, store, and analyze data.
- Azure Data Explorer is ideal for analyzing large volumes of diverse data from any data source, such as websites, applications, IoT devices, and more.
- This data is used for diagnostics, monitoring, reporting, machine learning, and additional analytics capabilities.
- Can combine with Microsoft Sentinel and Azure Monitor.




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# Case study



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# Use Case Scenarios (activity)

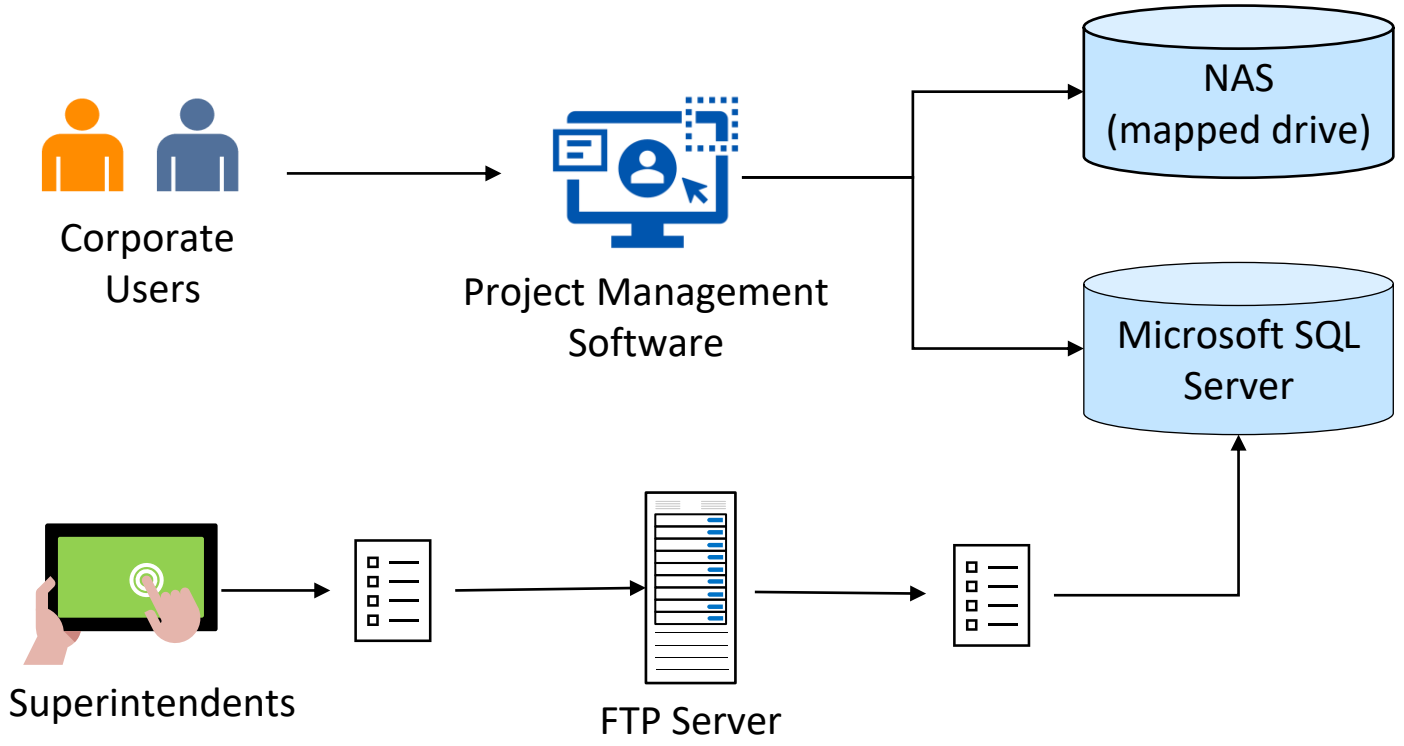


| | | |
|---|---|---|
| Azure Monitor | Measure user experience and analyze users' behavior for all external facing applications | Solution for collecting, analyzing, and acting on telemetry from your cloud and on-premises environments |
| Application Insights | You need to edit and run log queries | You need transaction diagnostics and performance statistics (client and server). |
| Azure workbooks | You need data analysis and the creation of rich visual reports | You need usage information on request rates, response times, and failure rates of an application |
| Log Analytics | | |

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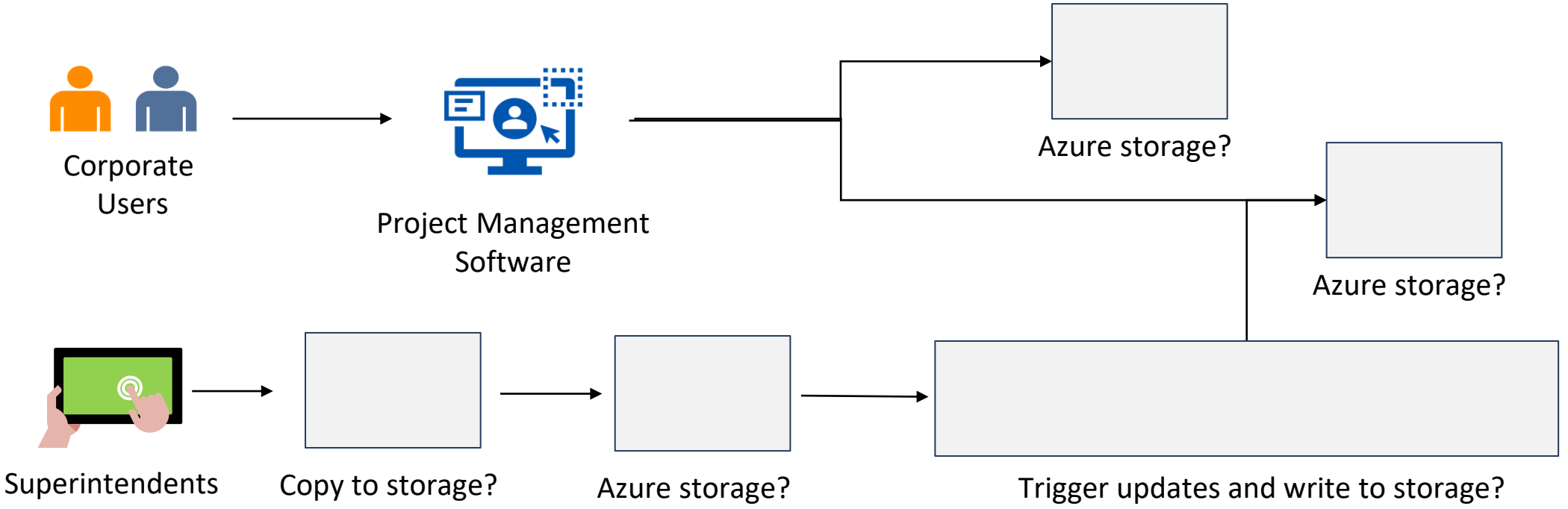
# Fabrikam Residences case study – PM software

- Third party Windows application
- 2 node cluster with SQL backend
- Images and documents stored on NAS device
- Corporate provides data on schedules and orders
- Superintendents record daily progress




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# Instructor solution - Fabrikam Residences PM software



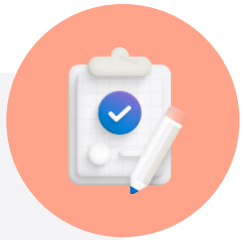




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# Learning Recap – Log and monitor solutions

Check your knowledge questions and review



**Reference modules**

- Monitor your Azure virtual machines with Azure Monitor
- Monitor app performance
- Describe monitoring tools in Azure

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# End of presentation



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